

Conjugated Systems & Aromaticity

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1 Conjugated Systems

1.1 Organic Acids & Bases Review

Core Concept 1.1: General Organic Compound Acidity

Memorizing the following pK_a values as baseline references is essential—they serve as anchors for estimating the acidities of related compounds by comparing functional groups, electron-withdrawing effects, and resonance stabilization of the conjugate base.

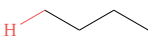
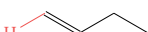
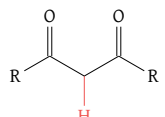
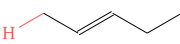
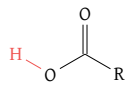
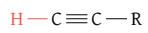
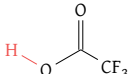
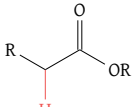
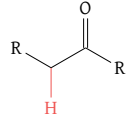
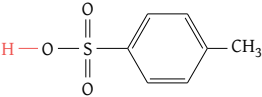
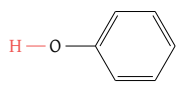
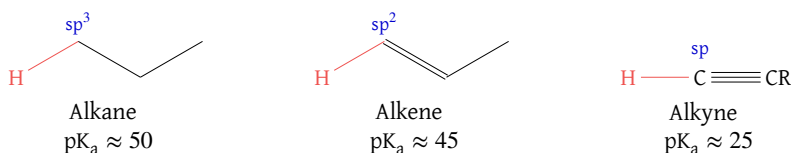
Compound	pK_a	Structure	Compound	pK_a	Structure
Alkane	50		Ammonium	10	
Alkene	45		α to Two Carbonyls	10	
Allylic	40		Carboxylic Acid	5	
Amine	35		Oxonium	0	
Alkyne	25		Trifluoro Acetic Acid	0	
α to an Ester	25		Hydrohalic Acid	<0	
α to a Ketone or Aldehyde	20		Sulfuric Acid	<0	
Alcohol (ROH)	15		Toluenesulfonic Acid	<0	
Phenol (PhOH)	10				

Table 1: Common acids and their pK_a values, ordered from least to most acidic

Core Concept 1.2: Acidity of Alkanes, Alkenes, and Alkynes

We now examine the pK_a values trend of alkanes, alkenes, and alkynes. The linear pattern observed can be rationalized by considering the following factors.

Figure 1: Alkane, alkene, and alkyne pK_a values

- Electronegativity of atom attached to the acidic H:** This factor cannot account for the trend, since the acidic hydrogen is attached to a carbon atom in all three cases, precluding any difference in electronegativity.
- Inductive Effect:** This factor is likewise insufficient, as only C–H bonds are present in the vicinity of the acidic proton.
- Resonance:** No resonance stabilization of the conjugate base is possible, so this factor does not explain the trend.
- Charge:** All species are neutral, so this factor offers no explanation.
- Hybridization:** Here a genuine difference emerges, as the acidic carbon is sp^3 -, sp^2 -, or sp -hybridized in the alkane, alkene, and alkyne, respectively. The lone pair generated upon deprotonation is stabilized most effectively in the sp -hybridized orbital, since its greater s character holds the electrons closer to the nucleus.

When the acidity cannot be rationalized from the structure of the acid itself, it is useful to instead examine the **conjugate bases**.

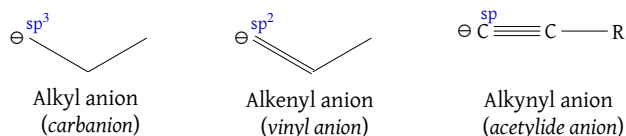


Figure 2: Conjugate bases of an alkane (left), alkene (middle), and alkyne (right)

Among the anions shown in Figure 2, the **alkynyl anion is the weakest base** because its lone pair is the most stabilized and thus least reactive. This arises because its e^- density is drawn more toward the carbon nucleus—away from the bonding region—making it less attractive toward H^+ relative to the alkenyl and alkyl anions. The following energy diagram below illustrates this comparison.

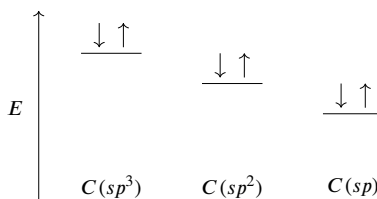


Figure 3: Diagram of sp^3 , sp^2 , and sp orbital energy levels; lone pair in the sp orbital is most stabilized because the relatively greater s character increases the probability e^- 's residing closer to the positive charge of the nucleus

Core Concept 1.3: K_{eq} and pK_a s

Any acid–base reaction has an equilibrium constant given by $K_{\text{eq}} = 10^{(\text{pK}_{a,\text{conj. acid}} - \text{pK}_{a,\text{acid}})}$. Using the pK_a values in Table 1, we can therefore estimate the K_{eq} of a given acid–base reaction to a reasonable degree of accuracy, as illustrated in the following examples.

(a) Protonation of ammonia by hydronium

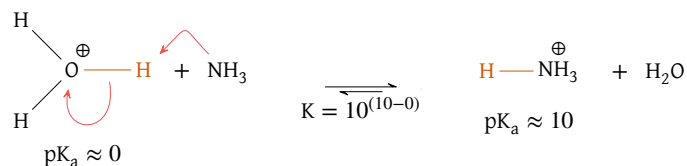


Figure 4: Protonation of ammonia by hydronium

(b) Enolate formation

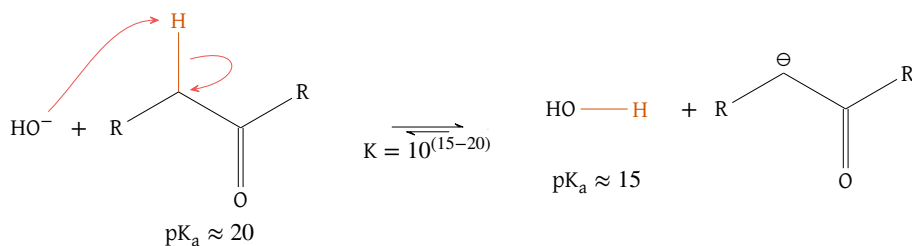


Figure 5: Enolate formation

(c) C-protonation of an enolate

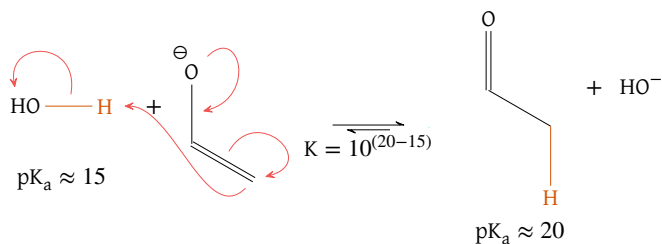


Figure 6: C-protonation of an enolate

(d) Enol formation

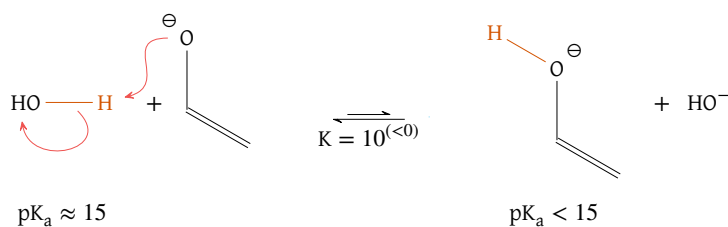
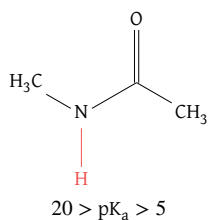


Figure 7: Enol formation

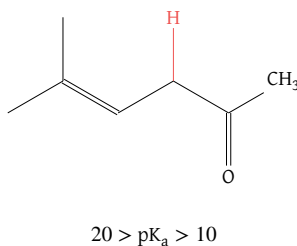
Furthermore, pK_a values of related structures can be predicted from the reference values in Table 1, as demonstrated in the following examples.

(a) Amine like α to carbonyl

Figure 8: Amine like α to carbonyl

The above molecule has a higher pK_a than a carboxylic acid ($pK_a = 5$), since nitrogen is less electronegative than oxygen, but a lower pK_a than a proton α to a carbonyl group ($pK_a = 20$), since nitrogen is more electronegative than carbon.

(b) α to carbonyl and allylic

Figure 9: α to carbonyl group and allylic

Here, The above molecule has a higher pK_a than a proton α to two carbonyl groups ($pK_a = 10$), but a lower pK_a than one α to a single carbonyl group ($pK_a = 20$), since it is additionally allylic.

1.2 Allylic Systems

Core Concept 1.4: Allylic Carbons

Allylic carbons are sp^3 hybridized carbons directly attached to a $C=C$ double bond, and an **allylic hydrogen** is one bonded to such a carbon.

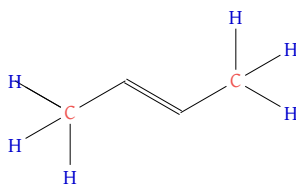


Figure 10: Allylic carbons (red) and allylic hydrogens (blue)

Core Concept 1.5: Allylic System Resonance

An allylic hydrogen exhibits greater acidity than both alkane and alkyne hydrogens, as resonance delocalizes the negative charge across the resulting allylic anion, thereby rendering the anion a comparatively weaker base.

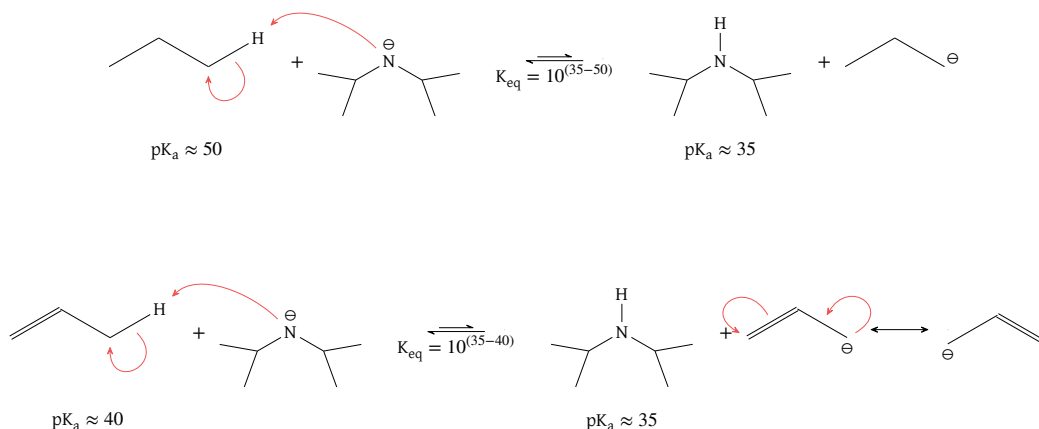


Figure 11: K_{eq} s of alkane (top) vs allylic (bottom) hydrogens; the K_{eq} of the allylic scenario is much more forward favoring because it is able to better stabilize the negative anion state charge

Just as negative charge (filled octets) is stabilized by resonance within allylic structures, radicals and positive charge (unfilled octets) are stabilized through this same resonance mechanism.

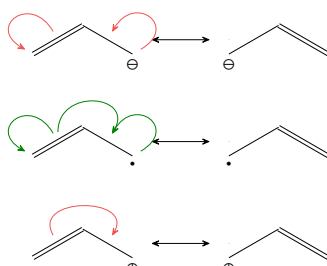


Figure 12: Filled octets (top), radicals (middle), and unfilled octets (bottom)

Critically, **any lone pair capable of participating in resonance must reside in a p orbital.**

Core Concept 1.6: Allylic System Bond Dissociation Energy

As a consequence of resonance-induced stabilization, the bond dissociation energies of distinct hydrogens within an alkene follow a predictable trend.

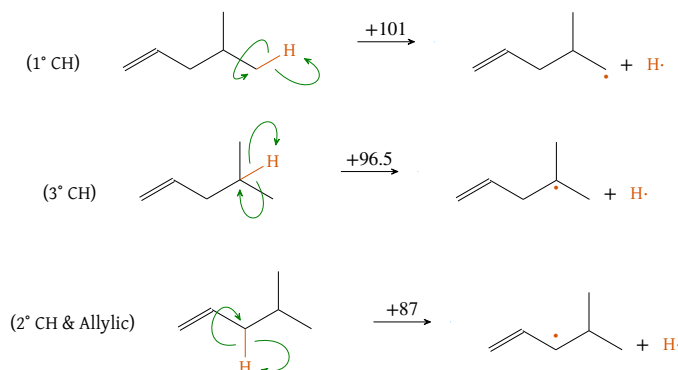


Figure 13: Primary (top), tertiary (middle), and secondary & allylic (bottom) hydrogen bond dissociation energies within the same alkene; secondary allylic hydrogen bond takes least energy to break due to resonance-induced stabilization

Above, the 3° carbon radical is lower in energy than the 1° carbon radical as a result of hyperconjugation between the radical carbon and neighboring σ bonds. The allylic radical, by contrast, receives comparatively less hyperconjugative stabilization as a 2° carbon, yet is stabilized through resonance, causing its corresponding bond dissociation to require the least energy of the three.

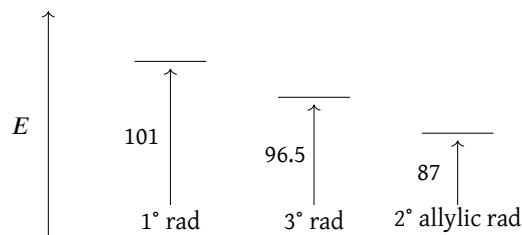


Figure 14: Energy diagram of Figure 13

This differential stabilization gives rise to chemical consequences that play a substantial role in practice, as illustrated by the reaction below.

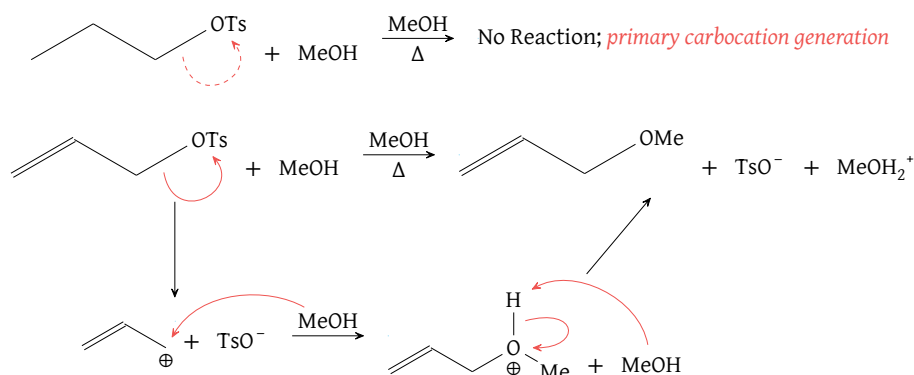


Figure 15: Non-allylic (top) vs allylic (bottom) leaving group primary carbocation formation viability; a primary carbocation can form in the bottom reaction because the cation is additionally allylic; Ts stands for a tosyl group, a common chemical group used in organic synthesis

Core Concept 1.7: Allylic System Molecular Orbital Structure

The molecular orbitals **relevant to allylic structures are the p orbitals**, whose energy levels can be compared in the context of sp - and sp^2 -hybridized carbons, as shown below. Notably, the energy level of sp^2 orbitals lies higher than that of sp orbitals.



Figure 16: Orbital energy level diagrams for sp and sp^2 hybridization

Superimposed on a skeletal diagram backbone, this orbital structure appears as follows.

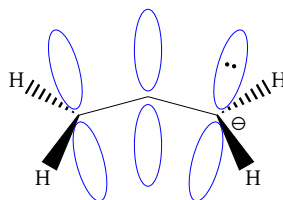


Figure 17: σ bond orbitals overlap along internuclear axis (handshake) whereas π bond orbitals overlap above and below internuclear axes. All carbons are sp^2 hybridized

Considered as a whole, the allylic p-orbital system produces an energy diagram with a distinctive three-level pattern, progressing from bonding to non-bonding to anti-bonding.

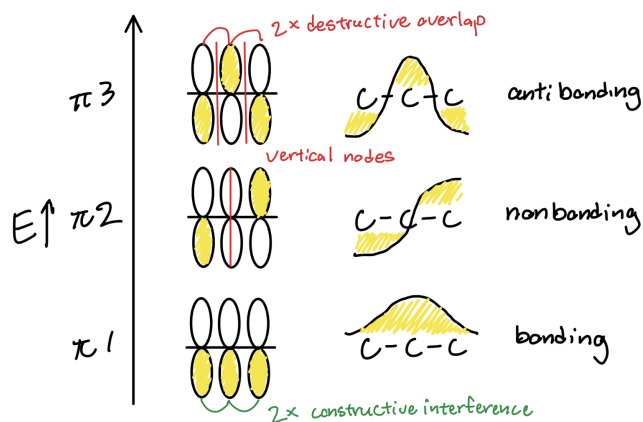


Figure 18: Energy diagram of allylic p-orbital system

Consequently, the HOMO and LUMO assignments differ among the allylic anion, radical, and cation.

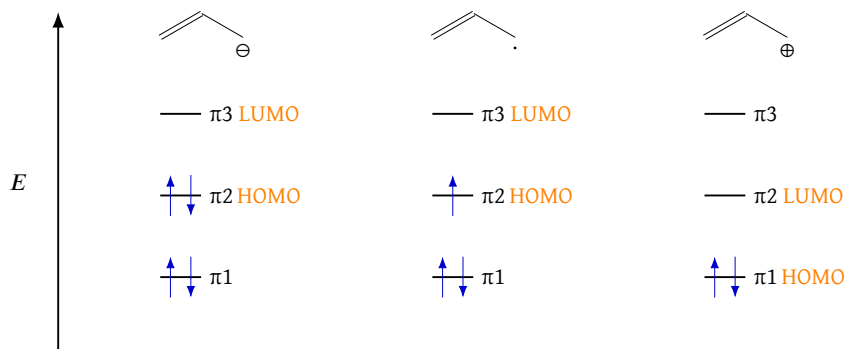


Figure 19: Molecular orbital diagrams for the allyl anion, radical, and cation, showing the filling of π_1 , π_2 , and π_3 with 4, 3, and 2 π e^- s respectively, along with HOMO LUMO assignments

This distinction is particularly relevant in the context of **frontier molecular orbital theory**, which holds that the HOMO of a nucleophile interacts with the LUMO of an electrophile (or vice versa) as explored in the context of diel alder reaction in UNCOMMENT AT END

1.3 Conjugated π Systems

Core Concept 1.8: Conjugated π System Stability

A **conjugated π system** refers to a network of adjacent allylic structures that permits resonance delocalization across a greater number of carbons. This effect can be observed by examining the hydrogenation enthalpies of the following alkenes.

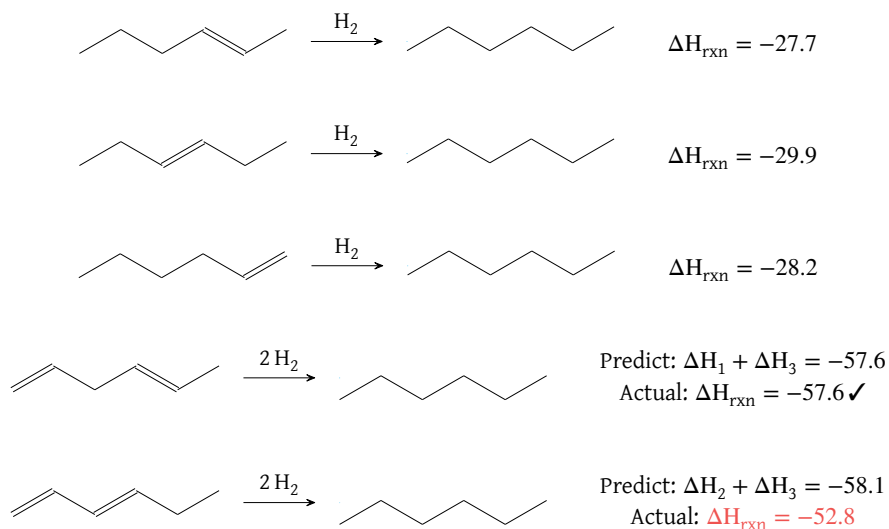


Figure 20: Enthalpies of alkene and alkadiene hydrogenations; the actual enthalpy of a conjugated π system alkadiene is lower than the predicted sum of individual bond hydrogenation enthalpies

Above, compound 5 exhibits greater stabilization than predicted by the simple additive sum of its constituent enthalpies, owing to the enhanced ability of conjugated π systems to stabilize excess or deficient charge.

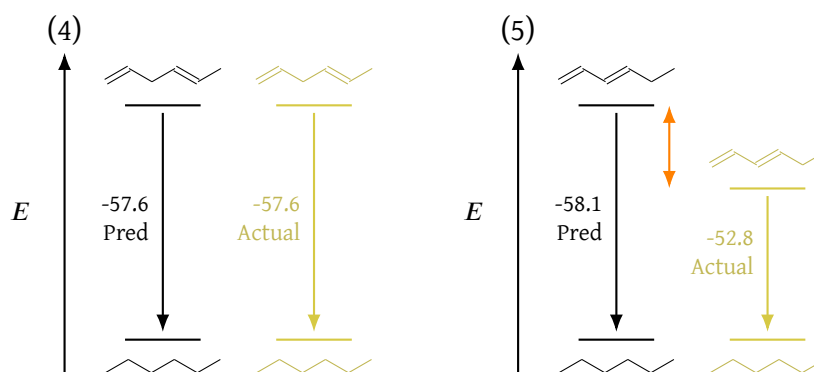


Figure 21: Comparison of predicted versus experimentally measured heats of hydrogenation for a non-conjugated diene (left) and a conjugated diene (right); the conjugated system's smaller actual value reflects a 5 kcal/mol stabilization arising from the four contiguous p -orbitals of the conjugated π system

A closer examination of the orbital energy diagrams for these conjugated π systems, following the same approach applied previously to the three-orbital π system (Figure 18), reveals that the conjugated system's orbital energy levels span an overall larger range while exhibiting smaller stepwise increases in energy between successive levels. Consequently, the HOMO–LUMO gap is comparatively smaller than that of the non-conjugated alkene π system.

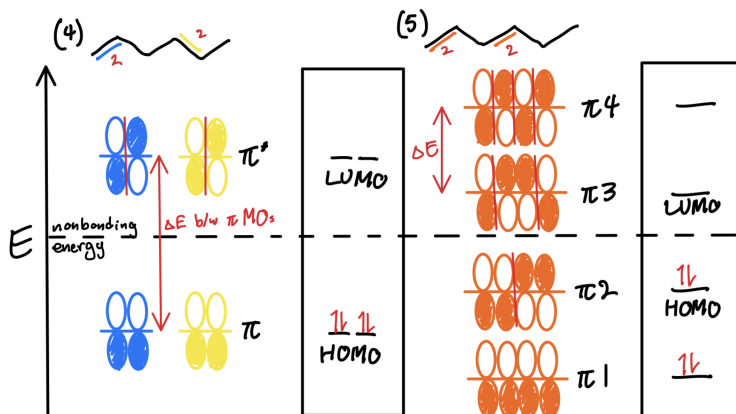


Figure 22: Energy diagrams of non-conjugated (left) and conjugated (right) π systems in alkadienes from Figure 21; source of stability in a conjugated π system are the lowest π MO e^- s

The governing principles for conjugated π system orbital energy diagrams are as follows.

- As conjugation increases (i.e., more p orbitals arranged in succession), the energy gap between π MOs decreases.
- As conjugation increases, the lowest-energy π MO decreases further in energy, while the highest-energy π MO increases further in energy.

It is important to recognize that every high-energy intermediate is stabilized to the greatest extent possible, and that resonance stabilization arising from conjugated π systems supersedes stabilization originating from hyperconjugation.

$$\text{resonance} > 3^\circ > 2^\circ > 1^\circ$$

This principle carries particularly significant consequences for the reactions covered within Alkenes & Alkynes, as many electrophilic additions proceed in either a Markovnikov or anti-Markovnikov manner, a distinction that depends on the relative stability of the intermediate state. Representative examples include alkene hydrohalogenation (Core Concept 1.9), which proceeds in a Markovnikov manner, and hydroboration-oxidation (Core Concept 1.13), which proceeds in an anti-Markovnikov manner.

Applying our understanding of conjugated π system stability, we now consider two illustrative examples.

Core Concept 1.9: Chlorination of Conjugated Cycloalkene

Our objective is to predict all products of the following reaction, including any relevant stereoisomers.

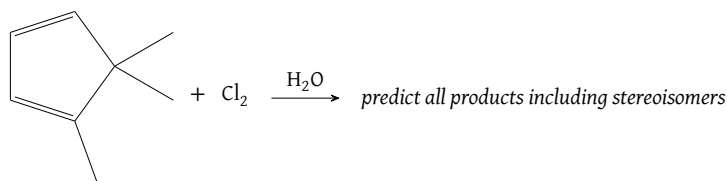


Figure 23: Cycloalkene chlorination reactants

First, we map out the chloronium intermediates.

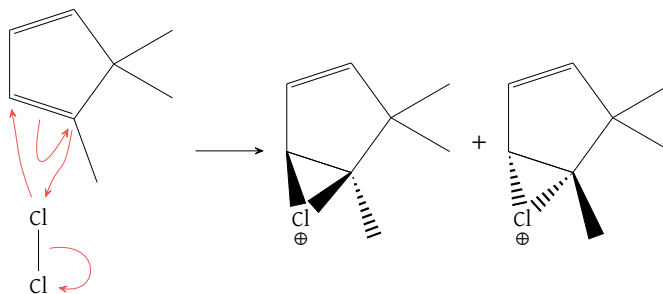


Figure 24: Top (*left*) and bottom (*back*) attacks leading to corresponding chloronium intermediates

Resonance is used to determine which carbon of the halonium intermediate is preferentially attacked by the weak nucleophile ROH.

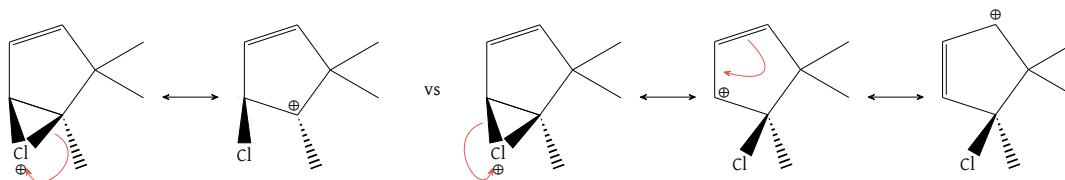


Figure 25: Tertiary "σ bond resonance" (*left*) vs secondary and allylic (*right*)

Because the allylic resonance contributor is more stable, the electrophilic carbons are consequently located at two distinct positions within the system.

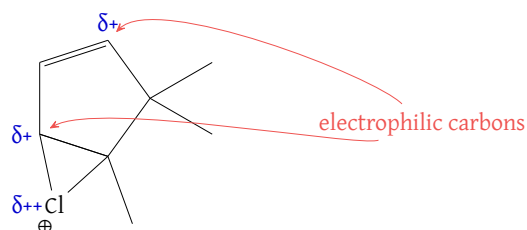


Figure 26: Electrophilic carbons on chloronium intermediate; *the top resonance contributor is the Cl*

Consequently, three possible modes of attack exist, giving rise to three distinct stereoisomeric products.

(a) **Tertiary electrophilic carbon attack**

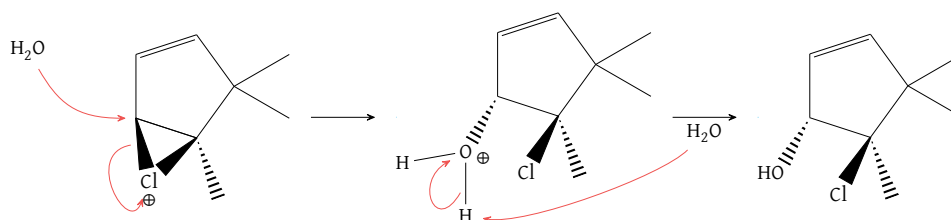


Figure 27: Tertiary electrophilic carbon attack

(b) **Secondary electrophilic carbon attack from front**

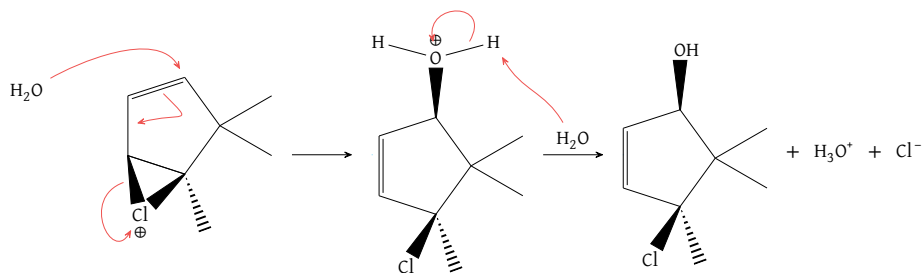


Figure 28: Secondary electrophilic carbon attack from front

(c) Secondary electrophilic carbon attack from back

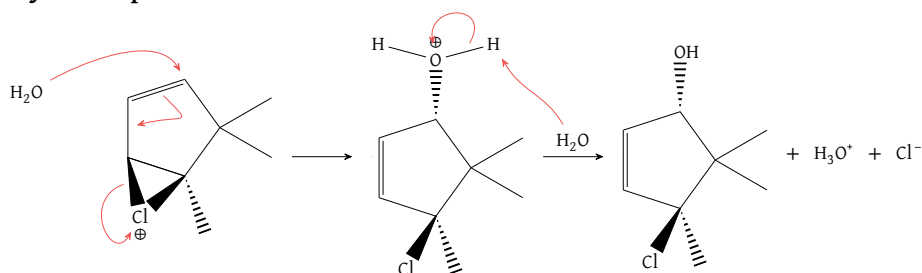
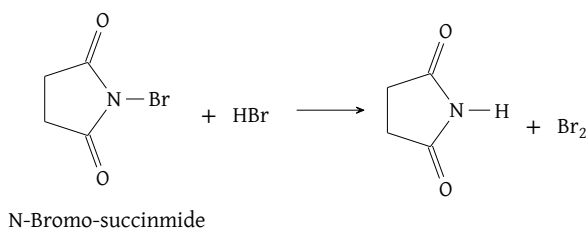


Figure 29: Secondary electrophilic carbon attack from back

Core Concept 1.10: Radical Bromination of Terminal Alkene

The radical bromination of a terminal alkene proceeds in a manner analogous to the reaction shown in Core Concept 1.9, wherein resonance creates the possibility of multiple distinct modes of attack and, consequently, three distinct products.

Br_2 is first generated in situ through **N-Bromosuccinimide (NBS)**, which enables the selective production of **monobrominated products**. It should be noted that direct addition of pure Br_2 to alkenes instead results in dihalogenation, as outlined in Core Concept 1.11.

Figure 30: N-Bromo-succinimide allows to the slow release of Br_2 to avoid alkenes adding Br_2 directly

Following the spontaneous dissociation of Br_2 into radical bromine atoms, the first propagation step occurs, generating a radical on the original alkene compound.

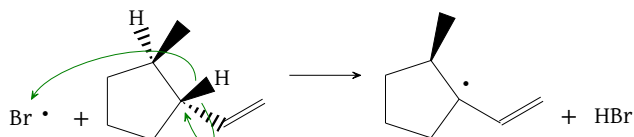


Figure 31: First propagation

This alkene radical and the remaining bromine radical can then undergo a second propagation step in three distinct manners.

(a) **Propagation 2, Tertiary Radical Attack**

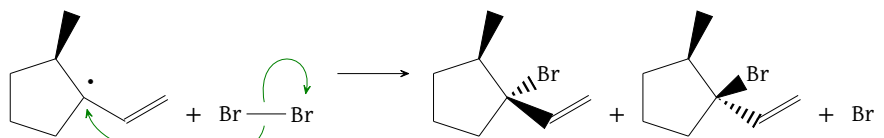


Figure 32: Propagation 2, tertiary radical attack

(b) **Propagation 2, Terminal Radical Attack**

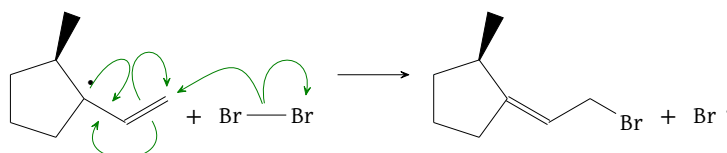


Figure 33: Propagation 2, terminal radical attack

(c) **Propagation 2, Rotated Terminal Radical Attack**

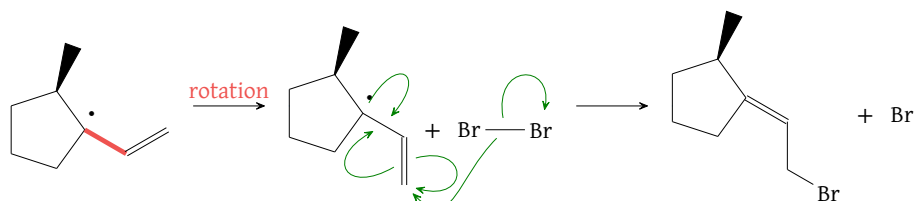


Figure 34: Propagation 2, rotated terminal radical attack

Taken together, the complete reaction proceeds as shown below.

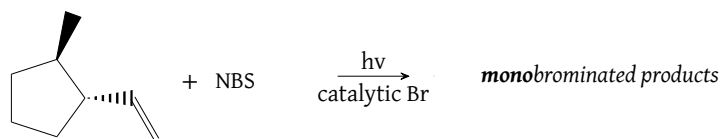


Figure 35: Radical bromination of terminal alkene full reaction